

Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

Next up:

- This week: finish Ch 5; Ch 6.1-2 and Kalman article for SVD
- Problem swap 2: solutions due tonight
- Quiz 10 on Friday (orthogonality, Gram–Schmidt)
- Problem session / office hours on Friday, 3-5pm, Ryerson 177
- HW₇ debrief/pre-brief on Sunday, 3-4 in Cobb 115
- HW₇ due next Monday
- Make a move on flex! And come by office hours!

Orthogonality fun facts!

The row space and nullspace of a matrix A are orthogonal subspaces.

(in fact, orthogonal complements)

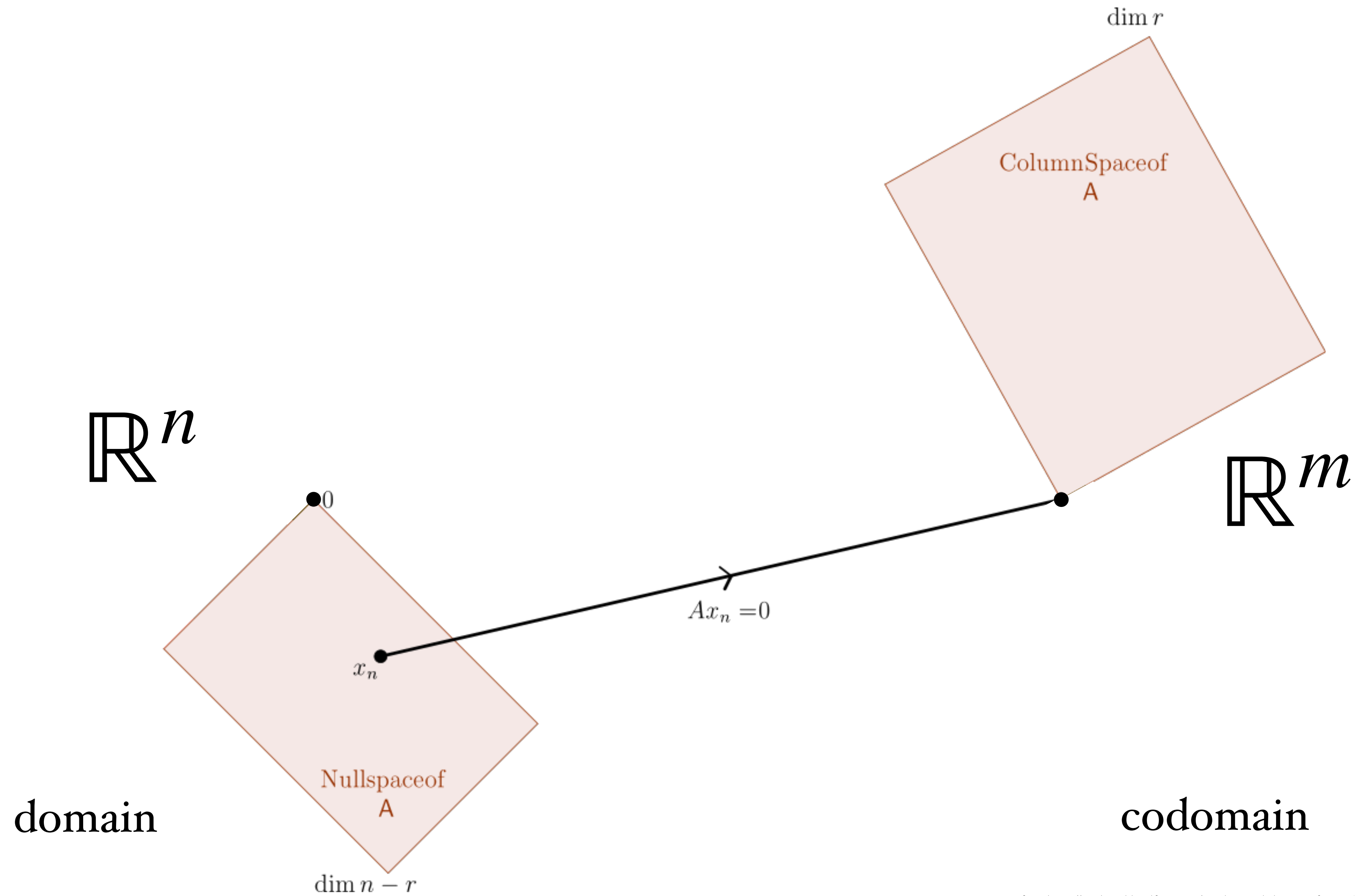
Same thing for the column space and the left null space.

For an orthogonal matrix A (one with columns an orthonormal basis)
we have $A^{-1} = A^T$.

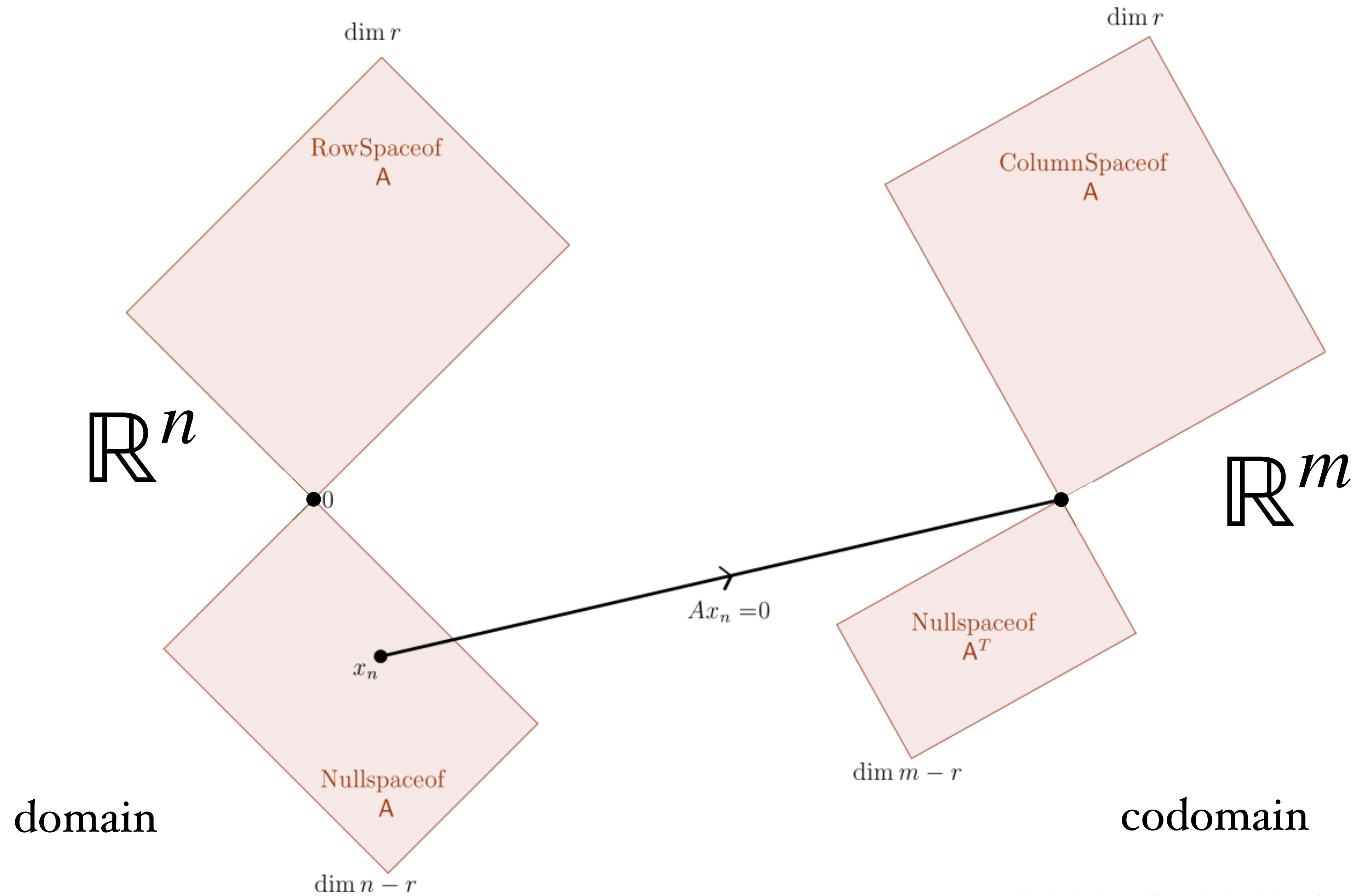
The determinant of a real orthogonal matrix A is either 1 or -1.

Note as special cases: every permutation matrix is orthogonal.

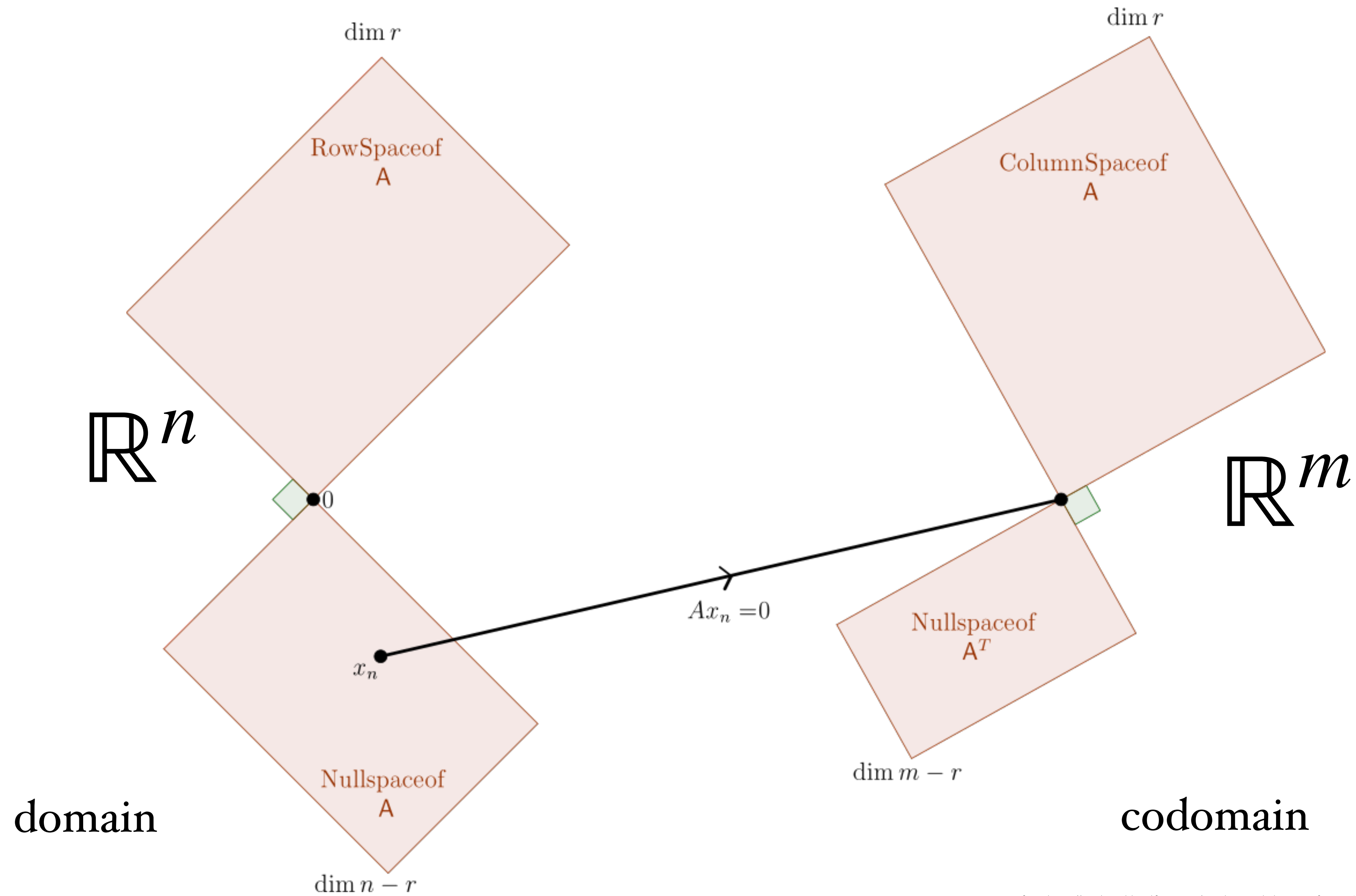
The fundamental theorem of linear algebra



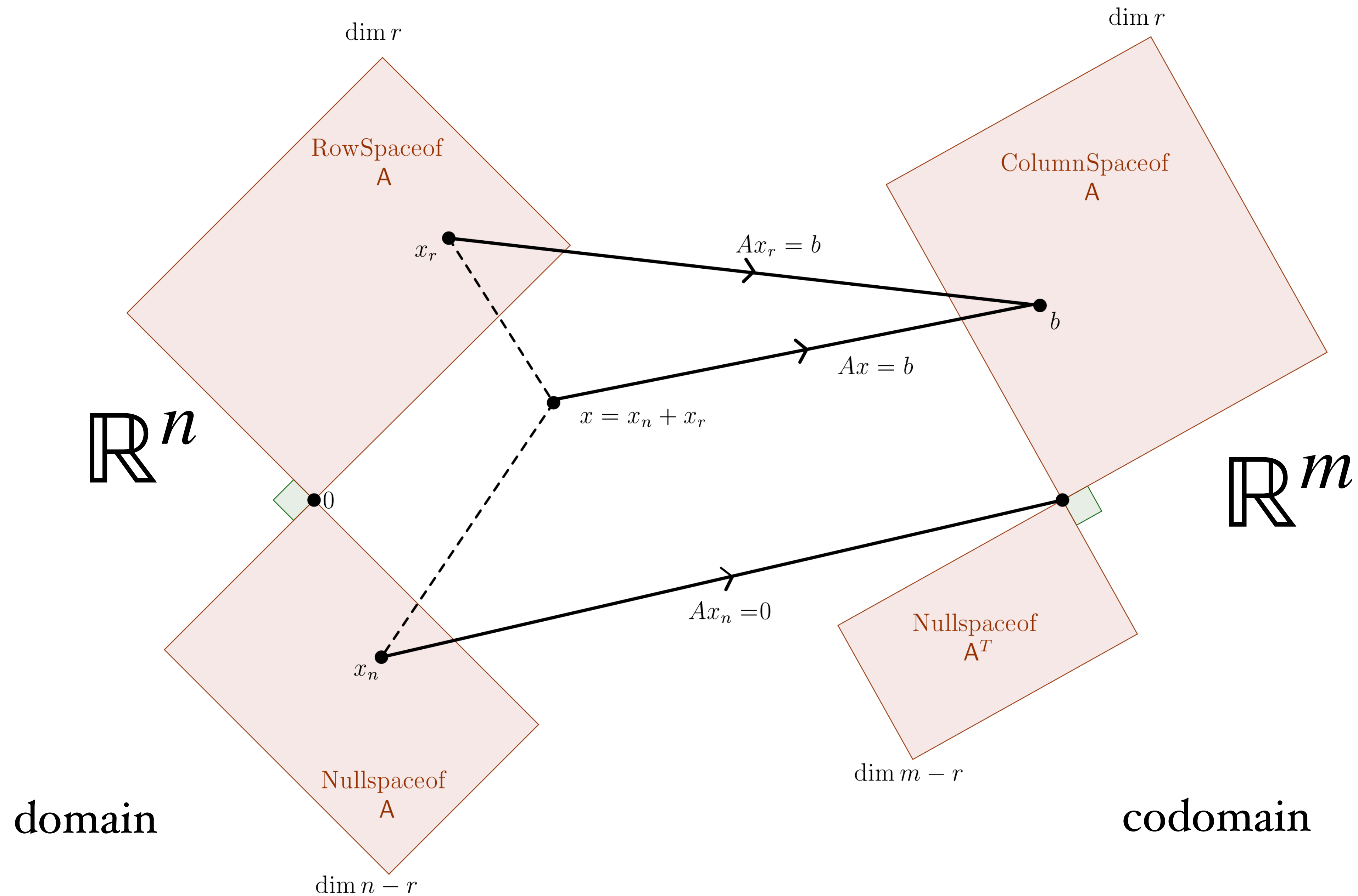
The fundamental theorem of linear algebra



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The fundamental theorem of linear algebra



Cauchy–Schwarz inequality (Theorem 5.1.7) Let $|\cdot|$ be the usual absolute value for real numbers and let $\cdot$ be ordinary multiplication of real numbers. For vectors x and y in an inner product space with inner product $(\cdot, \cdot)$ and norm $\|\cdot\|$, $|(x, y)| \leq \|x\| \cdot \|y\|$.

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Pf. Up to change of basis, every inner product is the dot product.

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$$\text{and so} \quad |x \cdot y| \leq \|x\| \|y\| \quad \text{since} \quad |\cos \theta| \leq 1$$

$\mathbb{R}$ $\mathbb{C}$

inner products are symmetric

$$\langle u, v \rangle = \langle v, u \rangle$$

The dot product is computed as

$$u^T v$$

Orthogonal matrices preserve the inner product:

columns are v_i and

$$v_i^T v_i = 1 \text{ and } v_i^T v_j = 0 \quad (i \neq j)$$

For orthogonal matrices we have

$$Q^{-1} = Q^T \text{ since } Q^T Q = I,$$

$$\text{and } \det Q = 1 \text{ or } -1$$

Symmetric matrices play a special role:

$$A = A^T$$

$\mathbb{R}$

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 $\mathbb{C}$

inner products are conjugate symmetric

$$\langle u, v \rangle = \overline{\langle v, u \rangle}$$

The dot product is computed as

$$\overline{u}^T v \quad \text{or} \quad u^* v$$

adjoint, or conjugate transpose

Unitary matrices preserve the inner product:

columns are v_i and

$$v_i^* v_i = 1 \text{ and } v_i^* v_j = 0 \quad (i \neq j)$$

For orthogonal matrices we have

$$U^{-1} = U^* \text{ since } U^* U = I,$$

and $\det U \in \mathbb{C}$ is on the unit circle

Self-adjoint (Hermitian) matrices play a special role:

$$A = A^*$$

Ch. 5 & 6: “Now with orthogonality!”

Row reduction:

apply row operations to make
an upper triangular matrix

LU decomposition:

$$A=LU$$

diagonalization:

$$A = PDP^{-1}$$

when A has n lin ind eigenvectors

matrix equivalence:

$$A= Q“I” P, \text{ with } Q \text{ \& } P \text{ invertible}$$

Gram–Schmidt:

apply column operations to make
an orthogonal matrix

QR decomposition:

$$A=QR$$

Spectral Theorem:

$$A = PDP^{-1}$$

with P unitary (orthogonal, for $\mathbb{R}$)

when A is self-adjoint (symmetric for $\mathbb{R}$)

$$A = A^*$$

Singular Value Decomposition

$$A= Q“D”P, \text{ with } Q \text{ \& } P \text{ orthogonal}$$

Upper triangular (Schur) representation (Thms. 6.1.1 & 6.1.2): Let A be the matrix of an operator on an inner product space. Then A can be represented as $A = UTU^*$, where U is a unitary matrix and T is an upper triangular matrix. In the case of real square matrices, any real square matrix A with all real eigenvalues can be represented as $T = UTU^{-1}$, where U is a real orthogonal matrix and T is a real upper triangular matrix.

Next up:

- This week: finish Ch 5; Ch 6.1-3 and Kalman article for SVD
- HW6 (written and WeBWorK) due tonight.
- HW6 debrief/prebrief today, 3:30-4:30 in Eckhart 207A;
No other office hours today.
- Problem swap 2: solutions due on Wednesday night
- Quiz 10 on Friday (orthogonality, Gram-Schmidt)
- Problem session / office hours on Friday, 3-5pm, Ryerson 177
- HW7 due next Monday
- Make a move on flex!